

Group 2

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Problem 2.10a. Let A be an element of SO_3 that represents a rotation with angle π . Describe the centralizer of A geometrically.

Proof. $A = \rho_{0,\pi} \in SO_3$. By the previous homework we know $\forall \alpha \in \text{Rig}(\mathbb{R}^3)$, $\alpha \rho_{0,\pi} \alpha^{-1} = \rho_{D(\alpha)(0),\pm\pi} = \rho_{D(\alpha)(0),\pi}$. $\forall \alpha \in Z(A)$, $\rho_{D(\alpha)(0),\pm\pi} = \rho_{D(\alpha)(0),\pi} = \rho_{0,\pi}$ so everything whose derivative fixes 0. Thus everything in $O(n)$ is the centralizer of the rotation by angle π $\square$

Problem 2.10b. Determine the centralizer of the reflection r about the e_1 -axis in the group M of isometries of the plane.

Proof. By previous homework, we know $\alpha r_L \alpha = r_{\alpha(L)}$. $\forall \alpha \in Z(r_{e_1})$, $\alpha r_{e_1} \alpha = r_{\alpha(e_1)} = r_{e_1}$. Thus α is anything which preserves the e_i axis which are any translation parallel to e_i , rotations around all points on the e_i axis by π , glide reflections across e_i . $\square$

Problem 2.13. Let N be a normal subgroup of a group G . Suppose that $|N| = 5$ and that $|G|$ is an odd integer. Prove that N is contained in the center of G .

Proof. Since N is closed under conjugation it is a union of conjugacy classes. $Z(G) \cap N$ is a subgroup of N . By Lagrange's Theorem, $|Z(G) \cap N| \mid |N| \implies |Z(G) \cap N| \mid 5$ since 5 is prime, $|Z(G) \cap N| = 1, 5$. $|Z(G) \cap N| = 5 \implies N \subset Z(G)$. If $|Z(G) \cap N| = 1$, then we have 4 elements of N which are not in the center. Since the conjugacy classes must be odd to divide $|G|$, only way to get 4 is $1 + 3$, but then $|Z(G) \cap N| = 2 \implies 2 \nmid 5$ which is a contradiction, thus $|Z(G) \cap N| \neq 1$. Thus $|Z(G) \cap N| = 5 \implies N \subseteq Z(G)$ $\square$

Problem 2.14a. The class equation of a group G is $1 + 4 + 5 + 5 + 5$. Does G have a subgroup of order 5? If so, is it a normal subgroup?

Proof. Let $g \in G$ be the element in the conjugacy class of 4 elements. Since $|G| = 20 = |Z(g)| |C_g| = 4 |Z(g)| \implies |Z(g)| = 5$. $Z(g)$ is a subgroup. $Z(g) = \mu_5$ and since it has prime order every element generates the subgroup and every nontrivial element has order 5. $\forall t$ in the conjugacy class with order 5, its centralizer must have order 4. Since all elements in $Z(g)$ have order greater than 4, they can't be t . Thus all nontrivial elements must be in the conjugacy class of 4. Thus the subgroup is a union of the trivial center and the conjugacy class with 4 elements. Thus it is a normal subgroup. $\square$

Problem 2.14a. The class equation of a group G is $1 + 4 + 5 + 5 + 5$. Does G have a subgroup of order 4? If so, is it a normal subgroup?

Proof. For t in the conjugacy class with 5 elements. $|Z(t)| = 4$. You can't form $Z(t)$ with unions of 2 or more conjugacy classes, one of which has to be the center. $Z(t) - Z(G) = 3$ all conjugacy classes which don't correspond to the center are greater than 3 so you can't take the union of any more. $\square$

For problem 16: Parts a - d are meant to address Keel's hint

Problem 2.16a. Explain that G acts on both G and G' "by conjugation" in such a way that $G \rightarrow G'$ is a map of G -sets (commutes with the G -action).

Show that for any x , $|Gf(x)|$ divides $|Gx|$, by constructing a natural surjective map $Gx \twoheadrightarrow Gf(x)$ and showing that any two fibres are in bijection.

Proof. Let $f : G \rightarrow G'$. G acts on G by conjugation. G acts on G' by conjugation after applying f , where $g(x) \rightarrow f(g)f(x)f(g)^{-1}$.

$\forall h \in G, g(f(h)) = f(g)f(h)f(g)^{-1} = f(ghg^{-1}) = f(g(h))$. Thus f is a homomorphism of G -sets.

$\forall a, b \in G \cdot f(x), \exists g \in G \ni a = g \cdot b$. $g \cdot f^*(a) = \{g \cdot h : f(h) = a\}$, Since f is a homomorphism of G -sets, $f_*(g \cdot f^*(a)) = g \cdot f_*(f^*(a)) = g \cdot a = b$. Thus $g \cdot f^*(a) \in f^*(b)$. Similarly $\exists g^{-1}$ s.t $a = g^{-1} \cdot g \cdot b$ or in other words, $b = g^{-1} \cdot a$. Thus there exists an inverse function $g^{-1} : f^*(b) \rightarrow f^*(a)$.

$\forall h \in f_*(a), g^{-1} \cdot g \cdot h = h$. Similarly $\forall h \in f_*(b), g \cdot g^{-1} \cdot h = h$. Thus there exists a bijection between the fibres.

Thus $|G \cdot x| = |f^*(a)||G \cdot f(x)|$. Thus $|G \cdot f(x)| \mid |G \cdot x|$ $\square$

Claim 2.16b. Show that G^x is a subgroup of $G^{f(x)}$

Proof. $\forall g \in G^x, g \cdot f(x) = f(gx) = f(x)$. Thus $g \in G^{f(x)}$. Thus $G^x \leq G^{f(x)}$ $\square$

Claim 2.16c. $|G \cdot x| = |X|, f : X \twoheadrightarrow Y \implies |G \cdot y| = |Y|$

Proof. $\forall y_1, y_2 \in Y, \exists x_1, x_2 \in X \ni f(x_1) = y_1, f(x_2) = y_2$. $|G \cdot x| = |X| \implies \exists g \in G \ni g \cdot x_1 = x_2 \implies f(g \cdot x_1) = f(x_2) \implies g \cdot f(x_1) = f(x_2) = g \cdot y_1 = y_2$. Thus there is only 1 orbit and G acts transitively. $\square$

Claim 2.16d. $X_x := f^{-1}(f(x)) \cong G^{f(x)}/G^x$

Proof. $G/G^x \cong G \cdot x = X$ and $G/G^{f(x)} \cong G \cdot y = Y$. We know there is a $f : X \twoheadrightarrow Y$. Let $\alpha : G/G^x \cong G \cdot x$ and $\beta : G/G^{f(x)} \cong G \cdot y$. There is a natural surjection between $G/G^x \cong G$ and $G/G^{f(x)}$ which is $\beta \circ f \circ \alpha$. Since $G^x \subseteq G^{f(x)}$ by the midterm problem 2 we know $X_x \cong G^{f(x)}/G^x$. $\square$

Problem 2.16e. Let $\phi : G \twoheadrightarrow G'$ be a surjective group homomorphism, let C denote the conjugacy class of an element x of G , and let C' denote the conjugacy class in G' of its image $\phi_*(x)$. Prove that ϕ maps C surjectively to C' , and that $|C'|$ divides $|C|$.

Proof. G acts on C and C' by conjugation. For $x \in C, \forall a \in C', \exists g \in G \ni a = g\phi(x)g^{-1} \implies a = \phi(gxg^{-1})$. Thus ϕ is a surjective homomorphism of G -sets. Conjugation acts transitively on conjugacy classes.

By claim 2.16a, $|C'| \mid |C|$. $\square$

Problem 2.17. Use the class equation to show that a group of order pq , with p and q prime, contains an element of order p

Proof. If $p = q \implies |G| = p^2$ thus any element in G has order 1, p , p^2 . If it has order p , we found it. If x has order p^2 then x^2 has order p .

If there is a conjugacy class of size q implies the centralizer of the element in the conjugacy class has order p implies it is a cyclic group of order p and every element in subgroup has order p .

If there is no conjugacy class of size q implies all conjugacy classes with elements not in the center have size p . The class equation is a sum of 1 and p . If $Z(G) = 1 \implies 1 + kp = pq \implies 1 = pq - kp \implies \gcd(pq, p) = 1$ which is a contradiction. Thus $Z(G) = p$ and thus all generators have order p .

If G is abelian. $\forall x \in G/\{1\}$. If x has order p , so we are done. If x has order pq then x^q has order p . If no x has order p or pq , it must have order q . Then every x has order q . $\forall g \in G, \langle g \rangle = \mu_q$. Define $H_g = \langle g \rangle$, $\prod_{g \in G} H_g$. The $\prod_{g \in G} H_g \rightarrow G$ is surjective. But $|\prod_{g \in G} H_g| = p^n$ and $|G| = pq$ and $pq \nmid p^n$. Thus a contradiction forms and not all elements can have order q . $\square$

Problem 2a. Note that for the action of G on itself by conjugation, the orbit $G \cdot g = [g]$, the conjugacy class of g , and the stabilizer $G^g = Z(g)$, the (subgroup of) elements in G that commute with g .

Now suppose $|G| = p^n$ for $n \geq 1$ and p prime. Explain why g IS NOT in the center iff $|Gg| = |[g]|$ is divisible by p . Now think about G partitioned into conjugacy classes.

Proof. If g not in the center, $|G \cdot g| \neq 1$. But since $|G \cdot g| \mid p^n$, $|G \cot g| = p^k$ for $k = 2 \dots n-1$. Thus $p \mid |G \cdot g|$.

If $p \mid |G \cdot g|$ then $|G \cdot g| \neq 1 \implies g \notin Z(G)$ $\square$

Problem 2b. number of singletons (conjugacy classes with just a single element) is congruent to 0 mod p . Conclude that the center is non-trivial.

Proof. Number of singletons = $Z(G)$. $|Z(G)| = 1, p^n$. If $|Z(G)| = 1$. Then $1 + (p^{a_1} + \dots + p^{a_n}) = p^k \implies p^k - (p^{a_1} + \dots + p^{a_n}) = 1 \implies \gcd(p^k, p^{a_1} + \dots + p^{a_n}) = 1$ which is not true and thus a contradiction occurs. Thus $|Z(G)| = p^k$ $\square$

Problem 3a. Let G be a group, and suppose that $G/Z(G)$ is cyclic. Prove G is Abelian (hint, lift a generator of the quotient).

Proof. Let G be a group, and suppose that $G/Z(G)$ is cyclic. Then the generator of $G/Z(G)$, $xZ(G)$. $(xZ(G))^m = x^m Z(G)$. $\forall a, b \in G$, $a \in x^k Z(G)$ and $b \in x^m Z(G)$, $a = x^k z_1$ and $b = x^m z_2$. $ab = x^k z_1 x^m z_2 = x^{m+k} z_1 z_2 = x^m z_1 x^k z_1 = ba$. Thus G is Abelian. $\square$

Problem 3b. Prove that if $|G| = p^2$, then G is Abelian.

Proof. The center is nontrivial. $|G/Z(G)|$ is a group with prime order and thus is cyclic and thus G is Abelian. $\square$